

Fully Interpretable Minimal Transformers: From Geometry to Algorithm

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1 SETUP

Abstract

We present a framework for building and interpreting minimal transformer models. By constraining a transformer's embedding dimension and head size to 22, we enable full two-dimensional visualization of its internal representations. Embeddings, query/key/value transforms, attention outputs, residual streams, and decision boundaries can all be seen directly. Our central claim is that the learned geometry implies an algorithm; the arrangement of points and boundaries in $\mathbb{R}^2$ can be read as a step-by-step procedure. We train a transformer on a simple task where it must produce the most recently observed even number whenever the '+' operator appears in a sequence of digits. Once trained, we visually walk through every step of the transformer's computation, and introduce a suite of interpretability visualizations that make the algorithmic interpretation of this procedure explicit. Our framework offers a pedagogical and experimental testbed to explore how transformers use informational geometry to implement next-token prediction.

Model Architecture

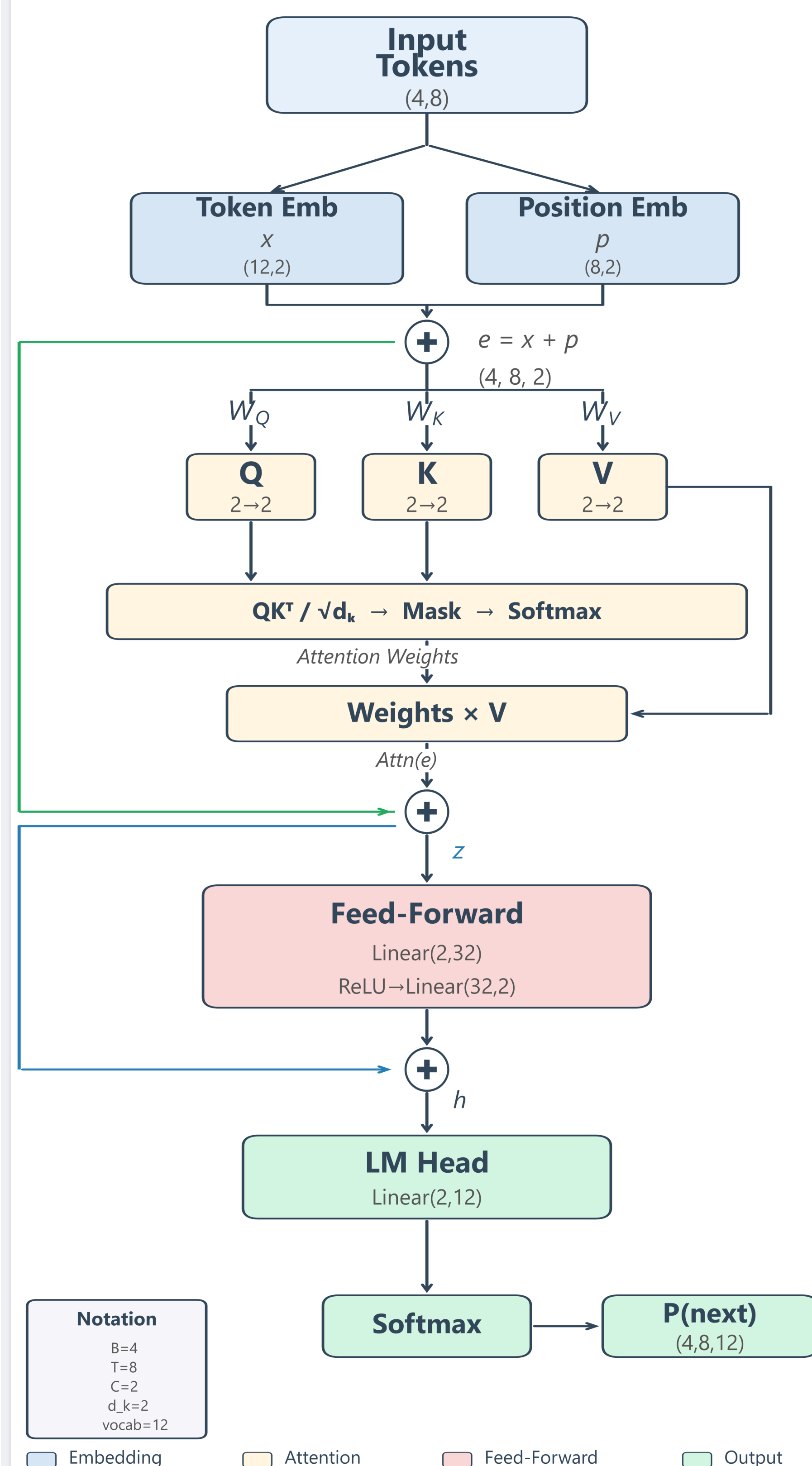


Figure 1. Architecture of the minimal transformer. Every component operates entirely in $\mathbb{R}^2$.

Training Data



Figure 2. Sample training sequences as heatmaps. Green: constrained position (immediately after '+') with the correct next token (the most recent even number). Gray: unconstrained position (any token is valid).

Learning Curve

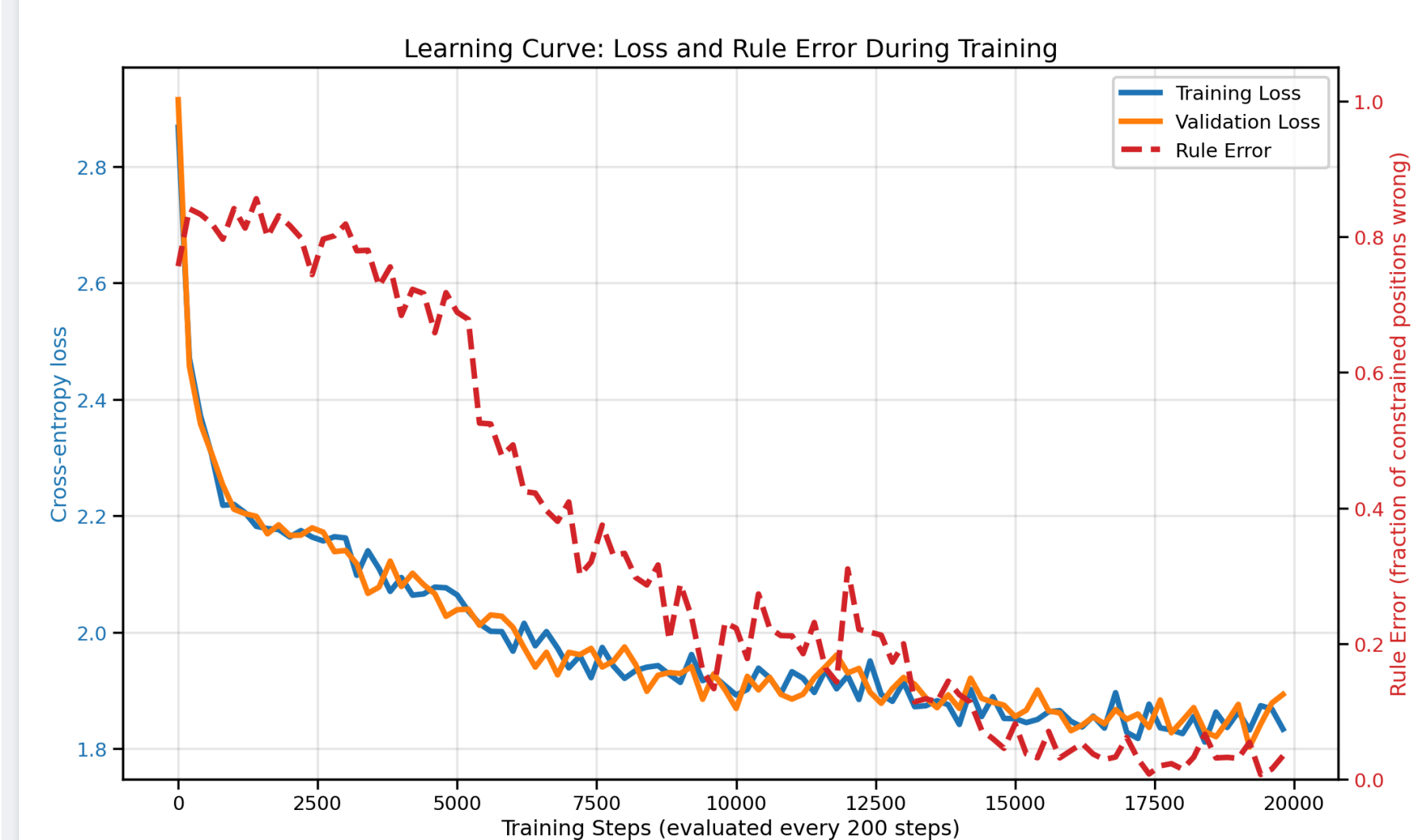


Figure 3. Training dynamics over 20,000 steps. Left axis: cross-entropy loss. Right axis: rule error — the fraction of constrained positions where the model's top predicted token is not the target. Rule error near 90% is chance; near 0% indicates the rule is learned.

2 INSIDE THE MODEL

Generated Sequences

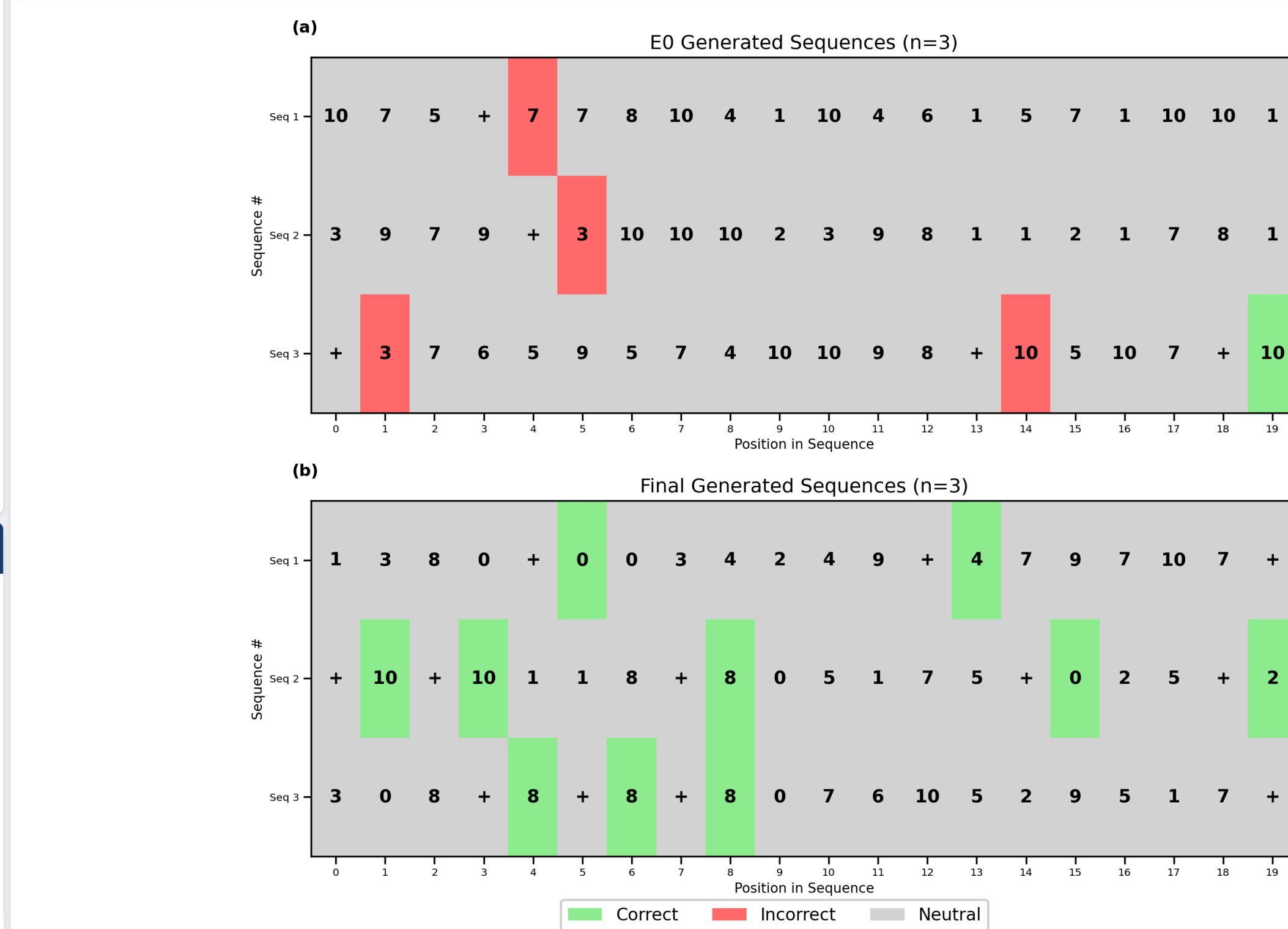


Figure 4. Model-generated sequences at initialization (top) vs. after training (bottom). Green = correct at constrained positions (after +), red = wrong, gray = unconstrained.

Learned Embeddings

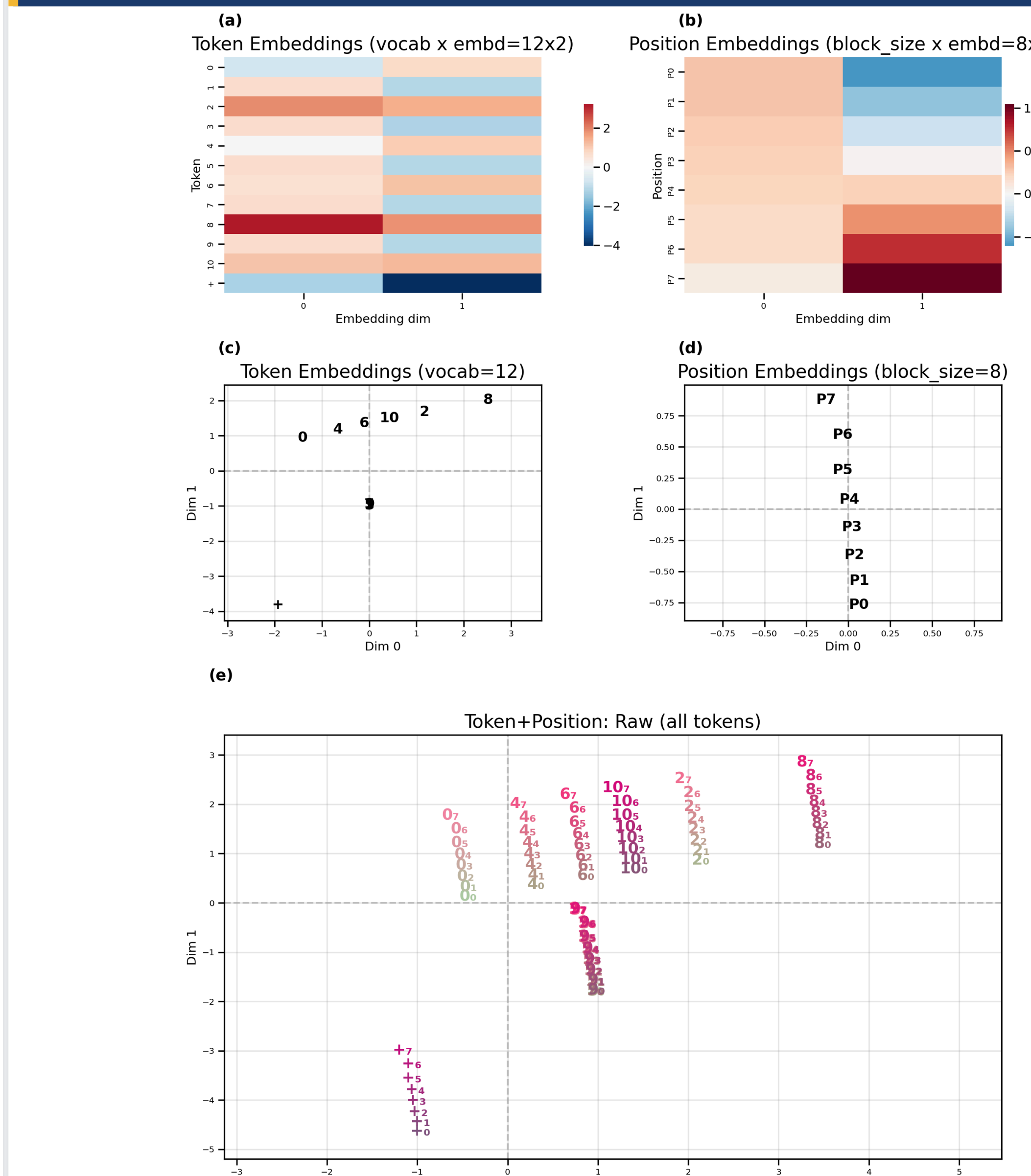


Figure 5. Learned embeddings. (a) Token embeddings x_i shown as heatmap. (b) Position embeddings p_i shown as heatmap. (c) Combined token+position embeddings shown as scatterplot. (d) Positions p_i shown as scatterplot. (e) Token+position embeddings e_i shown as scatterplot. Positions are indicated as subscripts next to each token e.g. $+_5$ is the + token at position 5.

Q, K, V Projections

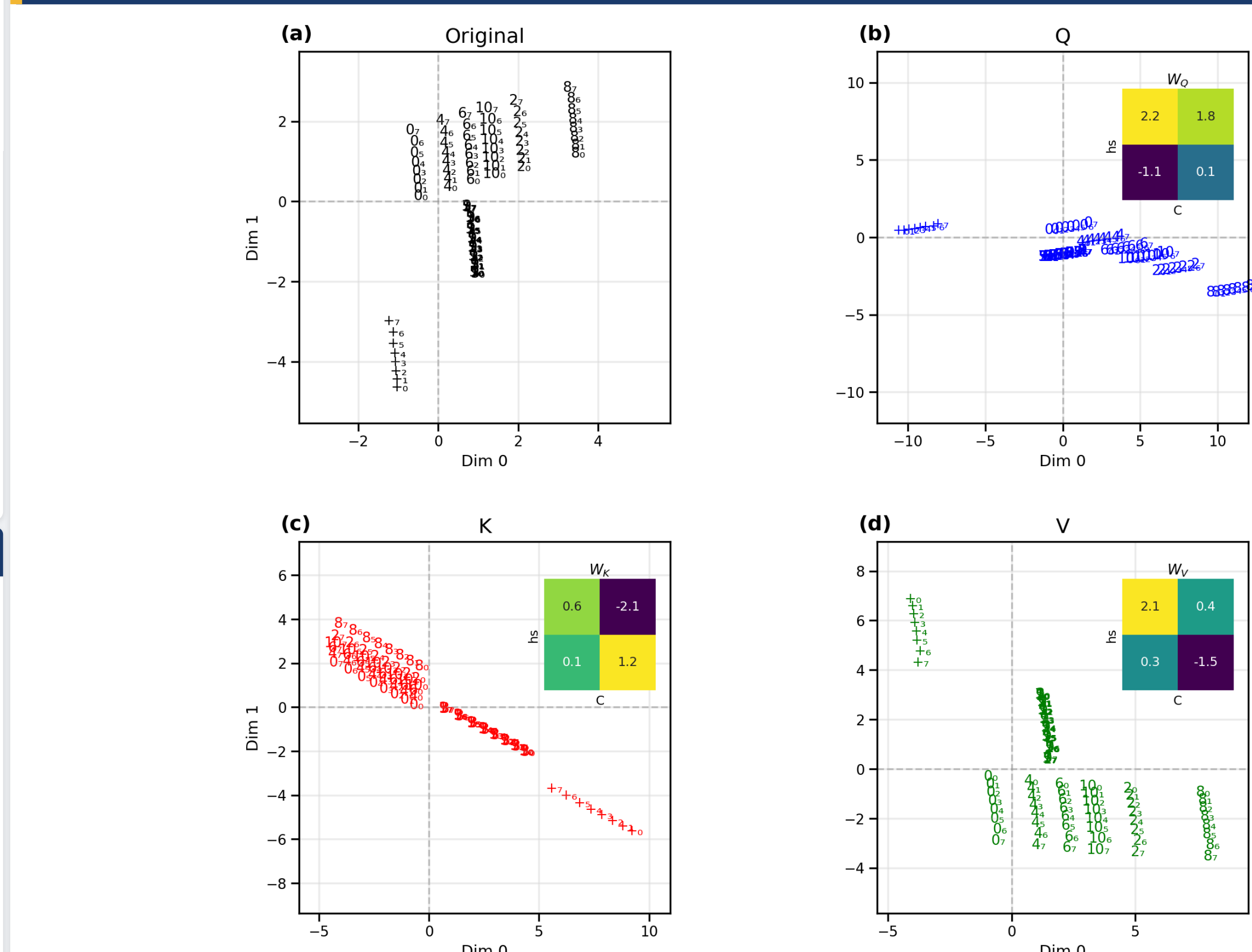


Figure 6. QKV projections. (a) Original combined token+position embeddings (as in Figure 5e). (b) Query-space (blue), with W_Q inset. (c) Key-space (red), with W_K inset. (d) Value-space (green), with W_V inset. All 96 token+position points in each panel.

Who Attends to Whom: Query–Key Geometry

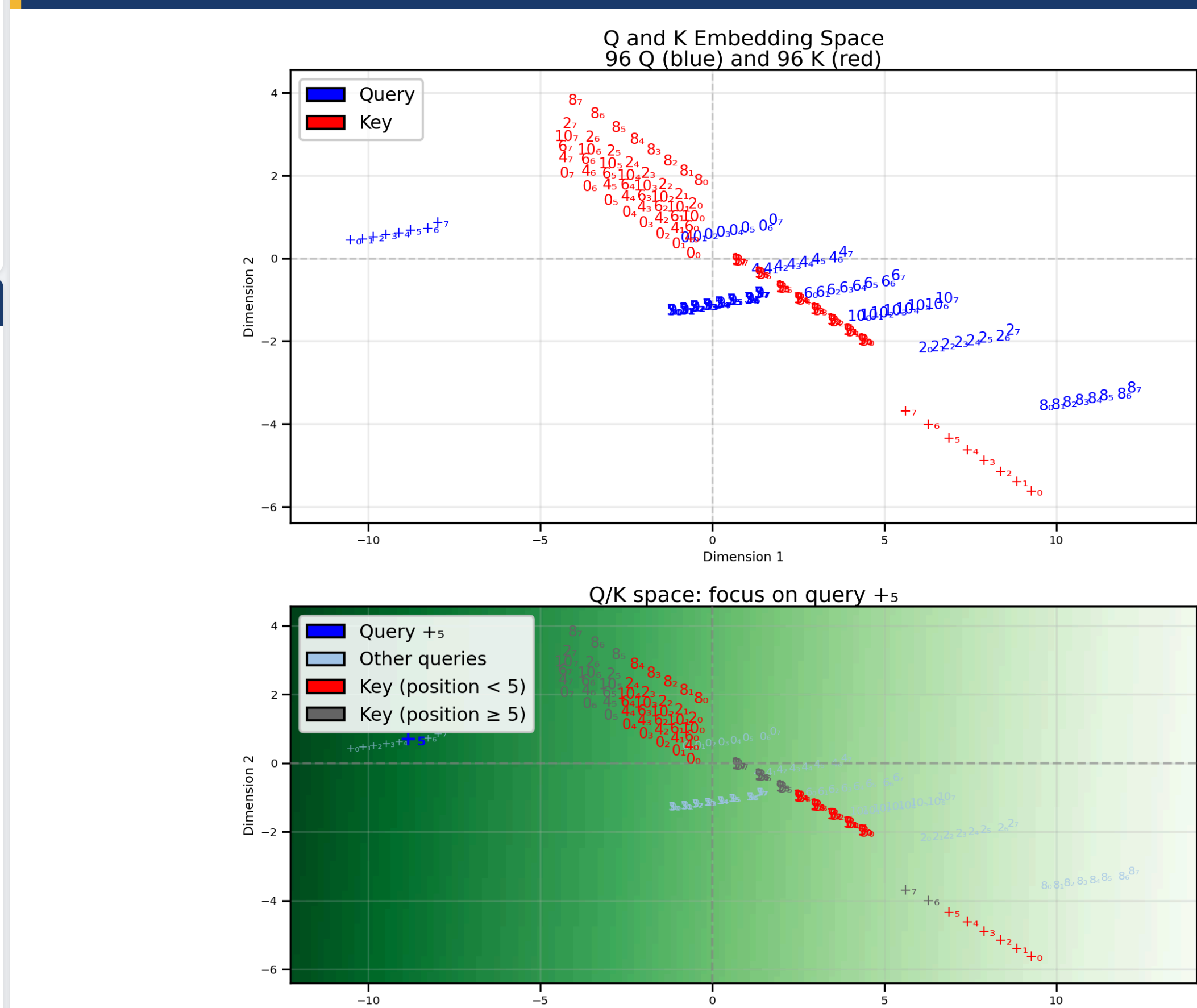


Figure 7. Query–key geometry in Q/K space. **Top**: joint query–key space (blue: queries; red: keys). **Bottom**: Focus on the query $+_5+5$ (blue); background color indicates the dot product of the $+_5+5$ query with every point in space. Keys with position ≥ 5 are grayed out due to the causal mask.

3 OUTPUT GEOMETRY

QK Scores for + Queries

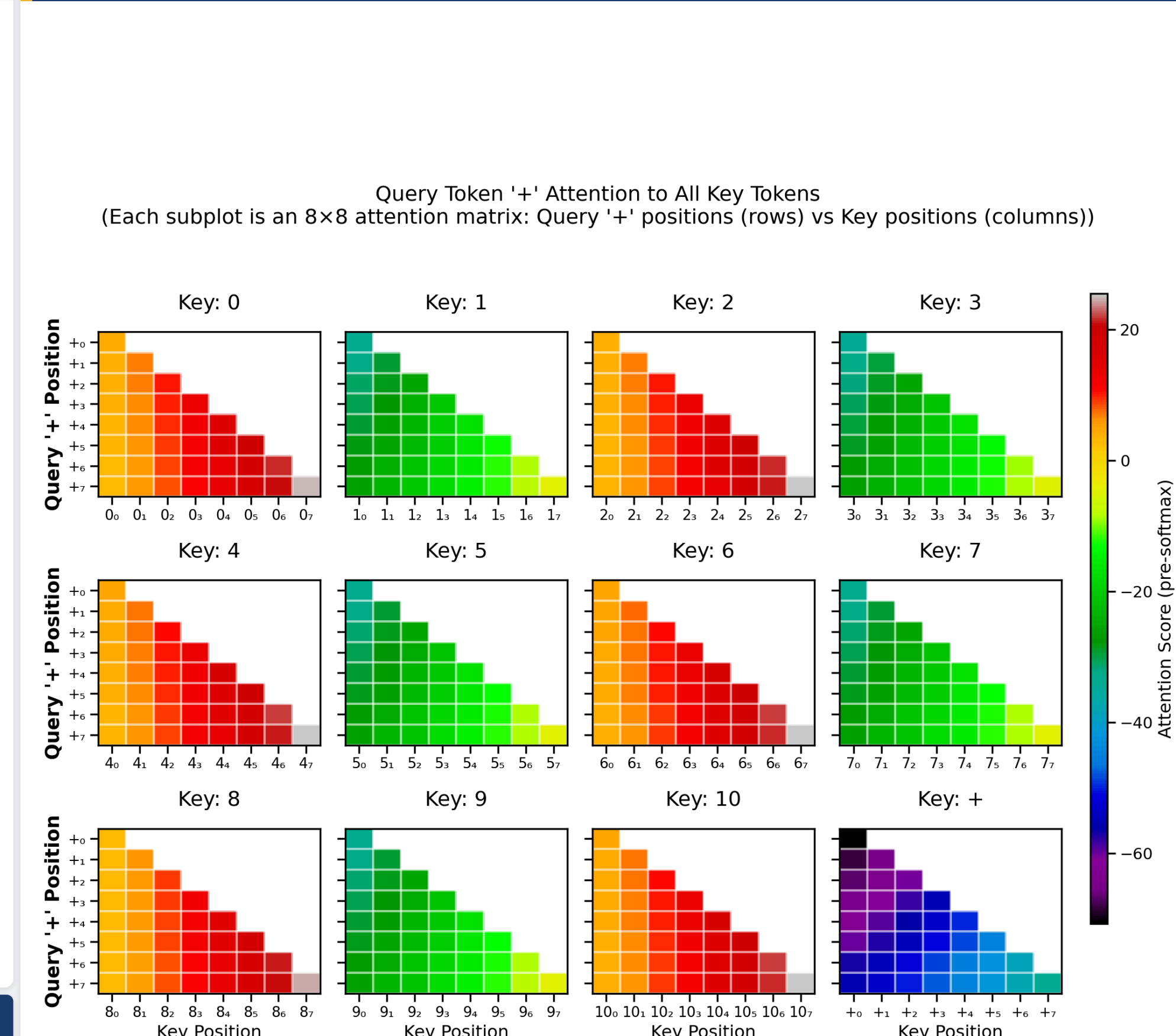


Figure 8. Pre-softmax query–key scores $QK^T QK^T$, restricted to + queries. Each subplot: an 8×8 matrix of dot products, + query positions (rows) vs. key positions (columns), for a particular key token.

Output Probability Landscape

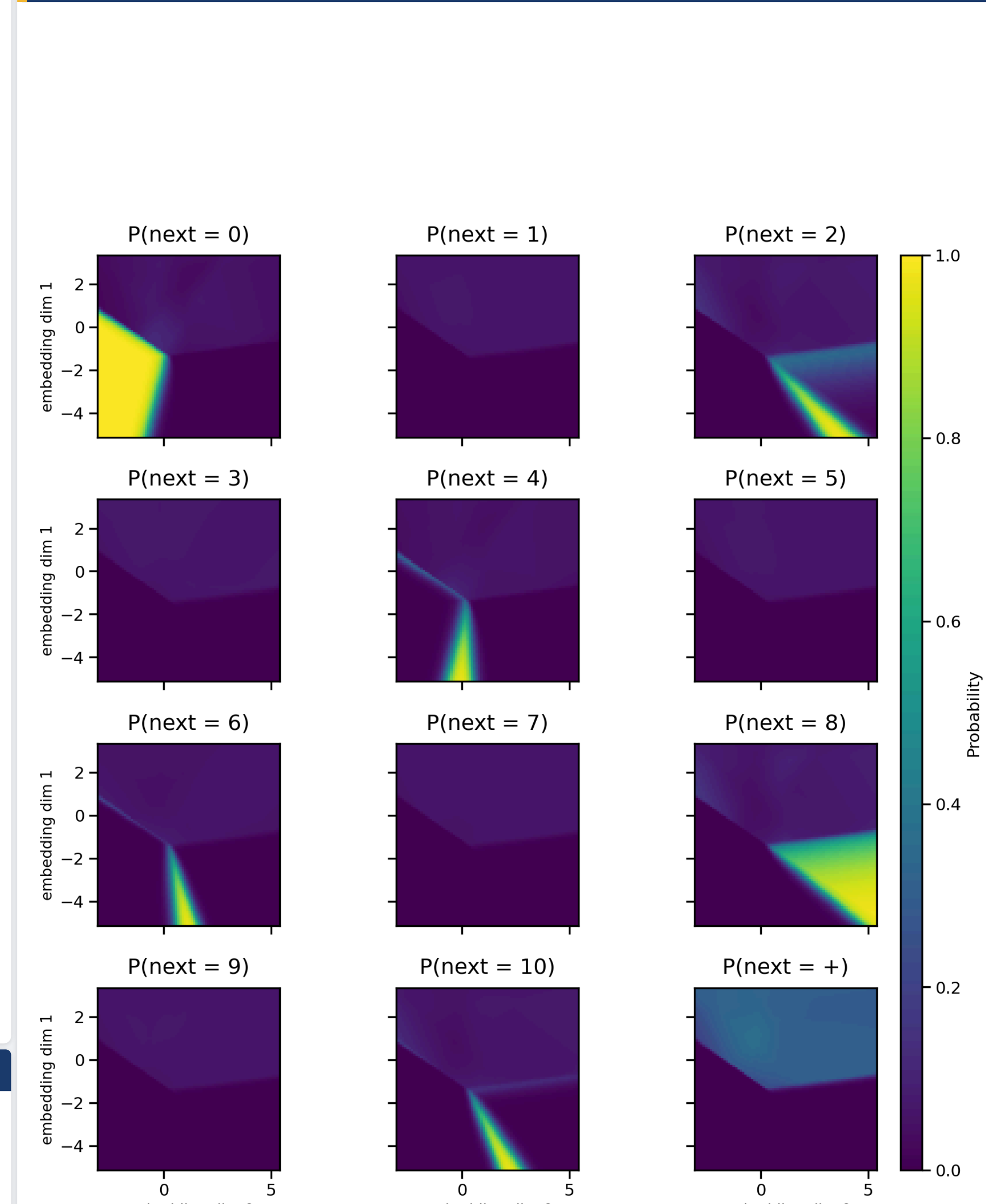


Figure 9. Output probability landscape (per-unit view). Each subplot shows the softmax probability $P(\text{next} = \text{token})P(\text{next} = \text{token})$ over the 2D plane for one output token (digits 0–10 and the + operator).

Output Landscape Summary

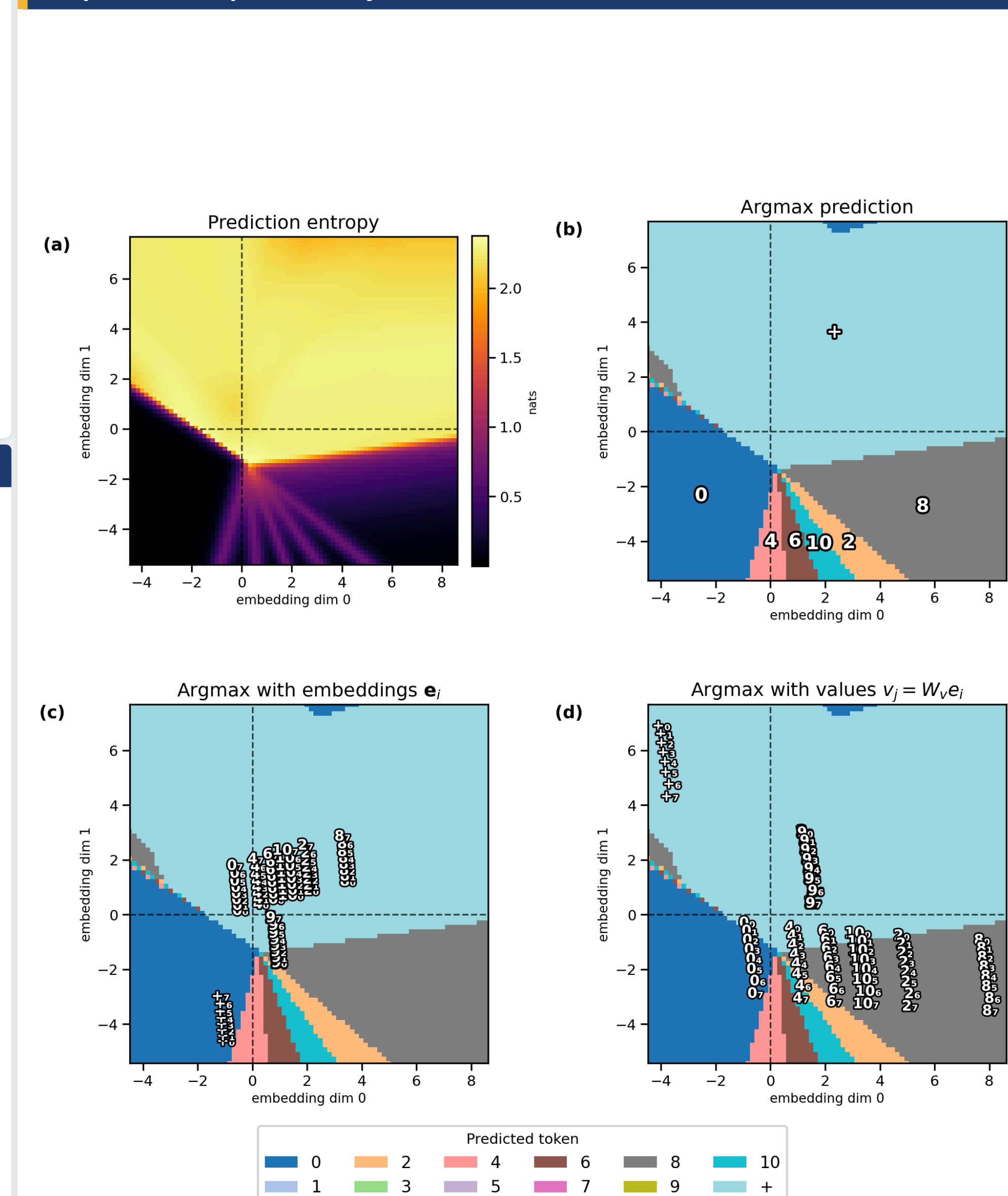


Figure 10. Output landscape summary. (a) Entropy of the output distribution (nats); high in the indifference region (top half), near zero in even-number target zones (bottom half). (b) Argmax prediction map: color and annotation indicate which token has the highest predicted probability at each point. (c) Argmax map with combined embeddings e_i overlaid. (d) Argmax map with value-transformed vectors v_i overlaid.